

MF20002.01 Investments

Group Project

Spring 2026

I. Basic information

- a. The deadline will be set in the eLearning system.
- b. Each team consists of 5 people.
- c. The group presentation will be held at the end of the course. Presentation order will be determined by a random draw.
- d. The final report consists of at most 10 pages, including appendices and references (12pt Times New Roman font, 1.5 line spacing).
- e. Choose project 1 or project 2.

II. Project 1 Guidelines

This project engages you in a "real life" asset allocation exercise. The project will require you to apply fundamental concepts of modern portfolio theory to achieve an efficient portfolio allocation to meet certain investment objectives. In short, the project will require you to implement what you learn in the first part of this course.

You are a portfolio manager at GBF College Fund. Your current investment policy consists of 60% stocks (spread 20/20/20 amongst large, medium, and small-cap stock portfolios, tracking indices of Shanghai Shenzhen CSI 300 Index (ticker:399300), CSI 500 index (ticker: 000905/399905), CSI 1000 index (ticker: 000852/399852)), 20% Corporate Bonds, and 20% Government Bonds (tracking indices of CSI Aggregate Bond Index (ticker: H11001) and SSE Government Bond Index (ticker: 000012)).

You believe the fund can achieve better performance by adding more asset classes. However, you need to provide solid and scientific evidence to convince the fund's board members. By adding more investment assets into the portfolio, your portfolio will gain benefits from diversification. You are interested in adding four more assets including Hong Kong Exchanges and Clearing Limited (HKEX) listed stocks, international stocks, real estate investment trusts, and commodities. You believe this new investment policy will outperform your current investment policy. You are considering assets tracking the Hang Seng Index (ticker: HSI), S&P 500 Index (ticker: SPX), Hang Seng REIT Index (HSREIT), and Gold ETF (ticker:518880).

To convince your board, you decided to conduct an empirical test with ten years of historical data from 2016 to 2025. You can download historical return data for the Shanghai Shenzhen CSI 300 Index (ticker:399300), CSI 500 Index (ticker: 000905/399905), CSI 1000 Index (ticker: 000852/399852), CSI Aggregate Bond Index (ticker: H11001), SSE Government Bond Index (ticker: 000012), Hang Seng Index (ticker: HSI), S&P 500 Index (ticker: SPX), Hang Seng REIT Index (HSREIT), and Gold ETF (ticker:518880) from China Stock Market and Accounting Research (CSMAR). For the risk-free rate, you may use the one-year deposit rate obtained from the CSMAR.

Your task is to:

1. Evaluate the performance of your current investment policy with 60% stocks and 40% bonds. Provide performance statistics, including the annualized mean returns, standard deviation, and Sharpe ratio of the portfolio.
2. Plot the portfolio frontier given only the five risky assets in which you are investing. Allow short sales when developing this frontier.
3. Is the portfolio of risky assets currently chosen by you an efficient portfolio? If not, calculate the investment proportions in the five assets chosen by the fund required to construct an efficient risky portfolio that would deliver the same expected return as the current choice of risky portfolio. For this part, assume there is no risk-free investment undertaken by the fund.
4. Calculate the investment proportions required to achieve the optimal (tangency) portfolio with the five risky assets, given the possibility of investing at a risk-free rate. Calculate the expected return and standard deviation of the tangency portfolio.
5. Continue to assume that you can invest in a risk-free asset. Calculate the investment proportions required to construct an efficient portfolio with an expected return equal to the existing portfolio's expected return. Also, calculate the expected standard deviation of returns on this portfolio.
6. Continue to assume that you can invest in a risk-free asset. Suggest an efficient portfolio allocation to achieve the college's objective of a "floor" rate of return equal to 8% per year. Also, calculate the expected standard deviation of such a portfolio.
7. You are interested in knowing if your portfolio should diversify its portfolio holdings by including real estate (represented by the Hang Seng REIT Index), HKEX listed stocks (represented by the Hang Seng Index), international stocks (represented by S&P 500 Index), and commodities (represented by Gold ETF) in its portfolio. Make a case for or against the inclusion of these alternative assets in the college's overall portfolio. Justify your decision by depicting the portfolio frontier in the presence of the nine risky assets vs. the earlier five risky assets. Ignore the risk-free asset in this part. Calculate the efficient portfolio allocations required to achieve the "floor" rate of return of 8% per year with this expanded universe of assets. Also, calculate the expected standard deviation of such a portfolio.

8. The calculations thus far make several assumptions regarding investment policy that are not allowed for the typical college fund. For this next part, assume that: a. You cannot take any short positions in risky assets. b. You cannot invest more than 30% of the total risky investment in any one asset category. Now calculate the efficient portfolio allocations required to achieve the “floor” rate of return of 8% per year. Also, calculate the expected standard deviation of such a portfolio.
9. Present a forceful argument for using your research results. Make a few succinct points that might help sway a board member skeptical of asset allocation methods. This member believes that individual stock selection is more important than asset allocation.

Notes:

1. Ignore the impact of forex return when constructing your portfolio. (e.g.: some assets are quoted in RMB, while other assets may be quoted in USD.)
2. You can use the total return index if it is available.
3. Links to downloading the data:
 - a. CSMAR (<https://data.csmar.com/>). It is only for academic purposes.

Additional Notes:

1. You can consult with me and the teaching assistant if you have questions.
2. Make sure that you cite relevant resources properly.
3. There will be a penalty if you miss the deadline.
4. You may use programming software (e.g.: Python) to finish this project.

Evaluation Criteria:

1. Data and empirical methods: 30%
2. Accuracy of construction the investment portfolio: 30%
3. Discussion of the main findings: 20%
4. Clarity of report: 20%

Project 2 Guidelines

If you are interested in quantitative investment and academic research, you may select a well-known academic paper on asset pricing, factor models, or fund performance evaluation from top finance journals or highly cited studies.

Your task:

1. Understand the research question, methodology, and key findings of the paper.
2. **Reproduce the main empirical results using real financial data.**
3. Extend the sample period to the most recent available time period.
4. Evaluate whether the original conclusions still hold out of sample (after the original sample period ends).
5. Submit a brief report that summarizes your work, results, and interpretation.

Notes:

1. You may use Python, R, or other standard statistical software.
2. Basic programming and data processing skills are needed.
3. Please confirm your chosen paper with me before you start the project.

Evaluation Criteria:

1. Understanding of the paper: 20%
2. Data and empirical methods: 30%
3. Accuracy of reproduction: 20%
4. Discussion of out-of-sample analysis: 20%
5. Clarity of report: 10%